a. more CO_2 in the blood
b. more O_2 in the blood
c. less CO_2 in the blood
d. less O_2 in the blood

12. Which of the following substances in tobacco smoke damage the gas exchange system?
a. carbon monoxide and carcinogens
b. carbon monoxide and nicotine
c. carcinogens and tar
d. nicotine and tar

13. Column I represents diseases and column II represents their symptoms. Choose the correctly paired option

Column I	Column II
(P) Asthma	(i) Recurring of bronchitis
(Q) Emphysema	(ii) Accumulation of W.B.C in alveolus
(R) Pneumonia	(iii) Allergy

a. P = iii, Q = ii, R = i
b. P = iii, Q = i, R = ii
c. P = ii, Q = iii, R = i
d. P = ii, Q = i, R = iii

14. Which of the following best describes the process of gas exchange in the lungs?
a. Air moves in and out of the alveoli during breathing.
b. Carbon dioxide diffuses from deoxygenated blood in capillaries into the alveolar air.
c. Oxygen and carbon dioxide diffuse down their concentration gradients between blood and alveolar air.
d. Oxygen diffuses from alveolar air into deoxygenated blood.

15. Make the correct pairs.

Columan–I	Column–II
(P) IC	i. maximum volume of air breathe in after forced.
(Q) EC	ii. Volume of air present after expiration in lungs.
(R) VC	iii. Volume of air inhaled after expiration.
(S) FRC	iv. Volume of air present after expiration in lungs.

(a) P – i , Q – ii , R – iii , S – iv
(b) P – ii , Q – iii , R – iv , S – i
(c) P – ii , Q – iii , R – i , S – iv
(d) P – iii , Q – iv , R – i , S – ii

16. Make the correct pairs.

Column–I	Column–II
(P) Tidal volume	i. 1000 to 1100 ml
(Q) Residual volume	ii. 500 ml
(R) Expiratory reserve volume	iii. 2500 to 3000 ml
(S) Inspiratory reserve volume	iv. 1100 to 1200 ml

(a) P – ii , Q – iv , R – i , S – iii
(b) P – iii , Q – ii , R – iv , S – i
(c) P – ii , Q – iv , R – iii , S – i
(d) P – iii , Q – iv , R – i , S – ii

17. Name the respiratory organs of flatworm, earthworm, fish, prawn, cockroach and cat.
18. Name the enzyme that catalyses the bicarbonate formation in RBCs.
19. Air moving from the nose to the trachea passes through a number of structures. List in order of the structures.
20. Which structure seals the larynx when we swallow?
21. Resistance in the airways is typically low. Why? Give two reasons.
22. How the body makes long–term adjustments when living in high altitude.
23. Why is pneumonia considered a dangerous disease?

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